

Chilled Water Cooling Coil Design Report - v2.1

Prepared by user: admin

Physical flow geometry: CROSS-FLOW (air perpendicular to tube/coolant direction). Water enters air-leaving side (cross-counterflow tendency)

Design Inputs

Input / construction	Value
Face W x H	1.200 x 0.850 m
Rows	6
Fin type	Wavy + louvers
Fin material	Aluminum
Tube material	Copper
FPI / fin pitch	10.0 1/in / 2.540 mm
Fin thickness	0.120 mm
Tube OD / wall	9.530 / 0.350 mm
Pt / PI	25.400 / 22.000 mm
Airflow	2.0400 m3/s = 7344 m3/h = 4323 CFM
Gross face velocity	2.000 m/s
Entering air DB / RH	27.00 degC / 50.0%
Coolant / concentration	Water / 0.0%
Coolant inlet / pressure	7.00 degC / 300.0 kPa abs
Coolant mass flow / circuits	1.5480 kg/s / 12
Water row progression	Water enters air-leaving side (cross-counterflow tendency)

Performance

Performance	Value
Total capacity	52.33 kW
Sensible / latent	36.08 / 16.25 kW
SHR	0.690
Leaving air DB / WB	12.08 / 11.73 degC
Leaving air RH	96.2%
Leaving coolant	15.80 degC
Condensate	23.12 kg/h
Wet fraction / state	83.7% / Mixed / partially wet by row
Air dP	10.4 Pa
Water dP avg	211.96 kPa
Water dP min / max	205.46 / 218.44 kPa
Target mode	Required leaving air condition
Target capacity	50.40 kW
Capacity margin	+1.93 kW
Target leaving air	DB 13.00 degC / WB 12.07 degC / RH 90.0%

Heat Transfer Diagnostics

Heat-transfer diagnostic	Value
C air	2.421 kW/K
C coolant	5.945 kW/K
Cmin / Cmax	2.421 / 5.945 kW/K
Capacity ratio Cr	0.407
Dry-reference NTU	2.657
Dry cross-flow effectiveness	0.774
Wet enthalpy effectiveness	0.673
Air temperature effectiveness	0.746
Air Re / Pr	2001 / 0.713
Coolant Re / Pr	8161 / 17.885
Capacity-rate limiting side	Air side
Resistance limiting side	Air side
Resistance split air / water / wall	80.7% / 19.2% / 0.1%

Velocities

Velocity / flow item	Value
Gross air face velocity	2.000 m/s
Max air velocity between fins/tubes	3.331 m/s
Coolant velocity in one tube	2.044 m/s
Supply header velocity	1.232 m/s
Return header velocity	1.232 m/s
Coolant mass flow per circuit	0.1290 kg/s

Calculated Geometry

Calculated geometry	Value
Tubes / row	33
Total tubes	198
Tube length	1.200 m
Fin count	472
Face / free-flow area	1.020 / 0.612 m ²
Free-area ratio	0.600
Air-side area	143.23 m ²
Inside tube area	6.59 m ²
Air-side correlation	Wang-Tsai-Lu wavy+louvered fin

Engineering Warnings

- More than 70% of the calculated dry thermal resistance is on the air side; increasing water flow alone is unlikely to give a large capacity gain.

Row-by-Row Thermal Marching

Rows are numbered in the air-flow direction. Physical geometry is CROSS-FLOW at every row. Water enters air-leaving side (cross-counterflow tendency). Converged: True in 23 iteration(s).

Row	Air DB in	Air DB out	Air WB out	RH out %	Water in	Water out	Q kW	Latent kW	Wet %	Mode
1	27.00	22.83	18.17	64.1	14.09	15.80	10.18	0.08	16.3	Partially wet
2	22.83	19.97	17.01	74.8	12.70	14.09	8.31	1.40	100.0	Partially wet
3	19.97	17.62	15.77	82.9	11.27	12.70	8.50	2.82	100.0	Fully wet
4	17.62	15.59	14.47	88.9	9.83	11.27	8.56	3.57	100.0	Fully wet
5	15.59	13.77	13.12	93.2	8.40	9.83	8.49	3.57	100.0	Fully wet
6	13.77	12.08	11.73	96.2	7.00	8.40	8.28	4.82	100.0	Fully wet

Method / Validation Note

- The physical fin-and-tube coil is cross-flow: air moves through the depth and coolant moves along the tubes, approximately 90 degrees to the air velocity.
- The water-row-progression option only changes which coil-depth side receives cold water. It is not a physical parallel-flow/counterflow change of the local air/tube orientation.
- Plain fins use the Wang, Chi & Chang (2000) plain fin j/f correlation. Wavy+louvered fins use the Wang-Tsai-Lu correlation as documented by ACHP.
- Wavy-only mode is deliberately labelled a calibration-required baseline rather than inventing unverified correlation coefficients.
- Water-side heat transfer uses Gnielinski with transition treatment; tube/header pressure loss uses Darcy-Weisbach with Churchill friction.
- Wet-coil capacity uses a segmented enthalpy-potential dry/part-wet/wet treatment. Exact tube-by-tube coolant temperatures require an explicit physical circuit routing map.
- Validate air HTC, wet air dP and fitting K values against actual coil test or trusted manufacturer data before production release.